External assessment 2022

Stimulus book

Digital Solutions

General instruction

- Work in this book will not be marked.

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Stimulus 1

This algorithm calculates the interest on a savings account based on the amount deposited and the number of years this amount remains in the account.

```
1  BEGIN
2   INPUT depositAmount
3   INPUT years
4
5   CALCULATE interestRate = calculateInterestRate(depositAmount)
6   CALCULATE savings = calculateSavings(depositAmount, years,
    interestRate)
7
8   OUTPUT ("For a deposit of ${depositAmount} for {years} year/s at an
    interest rate of ({interestRate} × 100)%, your total savings would be
    ${savings}.")
9  END
10
11 BEGIN calculateInterestRate (depositAmount)
12   IF depositAmount <= 10000
13      value = 0.04
14   ELSE IF depositAmount > 10000 AND depositAmount <= 50000 THEN
15      value = 0.03
16    ELSE
17      value = 0.02
18    ENDIF
19   ENDIF
20   RETURN value
21 END calculateInterestRate
22
23 BEGIN calculateSavings(depositAmount, years, interestRate)
24   FOR i = 1 TO years
25      deposit = depositAmount + depositAmount × interestRate
26   NEXT i
27   ENDFOR
28   RETURN deposit
29 END calculateSavings
```

Stimulus 2

These are sample procedures and programming structures for a one-time pad encryption algorithm.

Pseudocode sub-procedures

```
indexOf( <character>, <array> )
```

Purpose: To return the index of the located character in the array.

```
lengthOf( <string of characters> )
```

Purpose: To return the number of characters in a string.

```
checkLowerCase( <string of characters> )
```

Purpose: To return 1 if only lower case characters exist or return 0 otherwise.

```
SET
lowerCaseAlphabet ['a','b','c','d','e','f','g','h','i','j','k','l','m',
'n','o','p','q','r','s','t','u','v','w','x','y','z']
```

Example of usage: `lowerCaseAlphabet[4] is equal to 'e'`

Stimulus 3

This code library of functions enables secure access to student results in a database.

Function	`hashPassword`	
Description	Apply a hash function to a password	
Input		**Return**
• Plain text password		• Hashed value
Function call	`hashPassword(plaintext)`	

Function	`checkLogin`	
Description	Look up a student ID and hashed password in database	
Input		**Return**
• Student ID • Hashed value		• True/False
Function call	`checkLogin(studentID, password)`	

Function	`getResults`	
Description	Retrieve all results for a student ID	
Input		**Return**
• Student ID		• Array of line of encrypted data
Function call	`getResults(studentID)`	

Function	`decryptData`	
Description	Apply a decryption algorithm to the data using the supplied key	
Input		**Return**
• Key • Line of encrypted data		• Line of decrypted data
Function call	`decryptData(key, resultsLine)`	

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